Q1.

(a) An early type of echo sounder for a boat produces a sound of frequency 2000 Hz.

(i) The wavelength of this sound in water is 0.75 m. What is the speed of the sound in water? Give the unit.

.....

.....

1 mark

(ii) This sound is heard by the operator as a brief 'ping'. The speed of sound in air is 330 m/s. What is the wavelength of the sound in air? Give the unit.

.....

.....

1 mark

(iii) Modern echo sounders use higher frequencies than the early type. Complete the sentence below to describe how the wavelength produced by a modern echo sounder compares with that produced by the early type.

The wavelength produced by a modern echo sounder is

.....

1 mark

(b) The diagram shows a wave in a 'slinky' spring.



The frequency of the waves is 5 pulses per second and the wavelength is 80 cm.

What is the wave speed? Give the unit.

.....

1 mark
Maximum 4 marks

Q2.

Ultrasound waves are very high frequency sound waves. They cannot be heard by humans.

(a) Ultrasound waves can be used to clean jewellery.

The jewellery is put into a container of cleaning fluid.



Complete each sentence to explain how ultrasound can clean jewellery.

The ultrasound generator makes the molecules of the cleaning fluid

..... . The molecules knock particles of .....

from the surface of the jewellery.

(2)

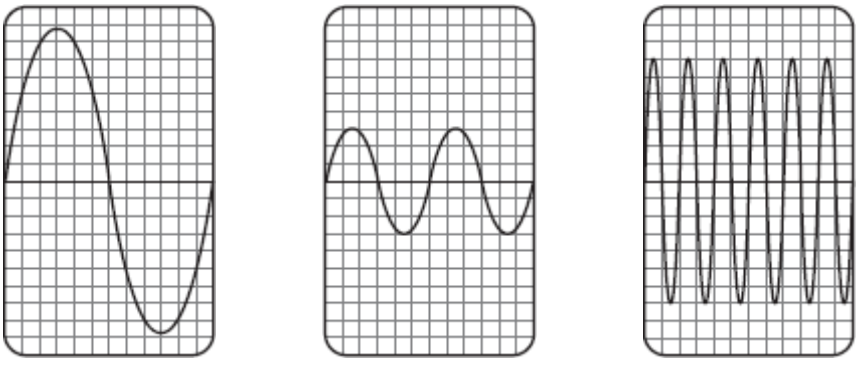
(b) Give a medical use for ultrasound.

.....

(1)

(c) Ultrasound waves can be represented on the screen of a cathode ray oscilloscope (CRO).

The diagrams show three ultrasound waves.
Each wave is represented on an identical CRO screen, **A**, **B** and **C**.



Screen **A** Screen **B** Screen **C**

(i) How many complete waves are shown on screen **B**? .....

(1)

(ii) Which screen shows the waves with the highest frequency?

Screen .....

(1)

(Total 5 marks)

Q1.

(a) (i) 1500 m/s **or** 1.5 km/s

unit required for the mark

accept '2000 × 0.75 m/s'

1

(ii) 0.165 m **or** 16.5 cm **or** 165 mm

unit required for the mark

accept '330 ÷ 2000 m'

1

(iii) shorter **or** less

*accept 'lower **or** 'smaller'*

1

(b) 4 m/s **or** 400 cm/s

unit required for the mark

*accept '5 × 0.8 m/s' **or** '5 × 80 cm/s'*

1

[4]

Q2.

(a) vibrate

*allow move more (vigorously) but **not** just move*

1

dirt / muck / grit / rust / dust etc.

*do **not** accept bacteria*

1

(b) any **one** medical use eg

ignore incorrect biological detail

- scanning unborn babies
- destroying (kidney) stones

1

(c) (i) 2

1

(ii) C

1

[5]